

Yale:

S&DS 265 / 565

Introductory Machine Learning

Societal Issues for Machine Learning

April 20

Yale

Housekeeping

- Assignment 5 due Thursday April 23 at midnight
- Final exam, Monday, May 4 at 2pm (cumulative, 3 hours, cheat sheet, practice exams posted to Canvas)
- Wednesday (last class): iML Survivor – Vote a topic off the final!
- Review sessions announced previously

Quiz 5

Ⓜ Average Score

89%

Ⓢ High Score

100%

Ⓣ Low Score

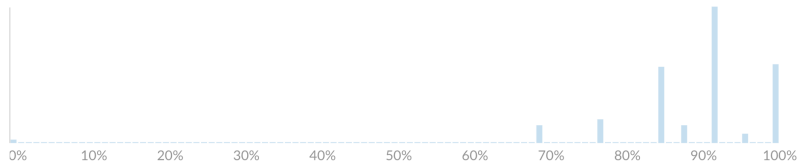
0%

Ⓢ Standard
Deviation

1.47

⌚ Average Time

18:51



Outline

- Recall: ML vs. AI
- Examples of bias
- Security/safety issues in LLMs
- Panel discussion

AI vs. ML

Machine learning focuses on making predictions and inferences from data.

AI combines machine learning components into a larger system that includes a decision making component.

An AI system exhibits a behavior, resulting from the collective decisions that are made.

Machine learning frameworks

- Supervised, unsupervised, semi-supervised
- Reinforcement learning
- Generative vs. discriminative models
- Representation learning

Example of representation learning:

Word embeddings

- Each word in vocab is mapped to 100 or 500 dimensional vector
- Based solely on co-occurrence statistics in corpus of text

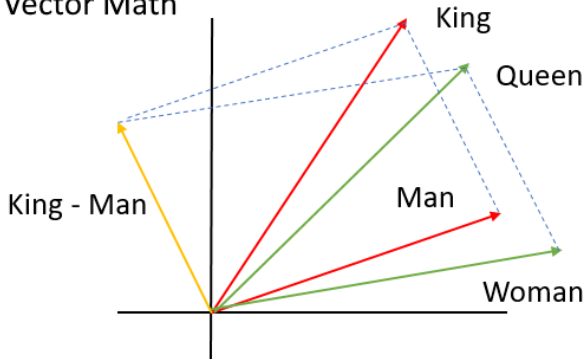
Example of representation learning: Word embeddings

Yale:

[0.78310001, 0.51717001, -0.38207 , -0.23722 , -0.31615999, 0.30805001, 0.76389998, 0.064106 , -0.74913001,
0.60585999, -0.23871 , -0.16876 , -0.25634 , 1.07270002, -0.29967999, 0.020095 , 0.54500997, -0.17847 , -0.26675999,
-0.11798 , -0.48692 , 0.22712 , 0.017473 , -0.4747 , 0.44861001, -0.084281 , -0.30412999, -1.13510001, -0.14869 , -0.11182 ,
-0.32530001, 1.0029 , -0.35742 , 0.35148999, -1.10679996, -0.064142 , -0.72284001, 0.14114 , -0.41247001, -0.16184001,
-0.54576999, -0.12958001, -0.88356 , -0.089722 , 0.10555 , -0.12288 , 0.92851001, 0.50032002, 0.1349 , 0.21457 ,
0.35073999, -0.73132998, 0.39633 , -0.43239999, -0.38815999, -1.34669995, 0.37463999, -0.79386002, 0.11185 , 0.18007 ,
-0.75142998, 0.24975 , -0.094948 , -0.36341 , 0.24869999, -0.22667 , 0.32289001, 1.29489994, 0.42658001, 1.29120004,
-0.13954 , 0.68976003, 0.21586999, 0.13715 , -1.00919998, 0.028827 , 0.11011 , -0.1912 , -0.073198 , -0.52449 , 0.49199 ,
0.14463 , -0.18844 , -0.75536001, -0.28704 , 0.019113 , 0.30349001, -0.74425 , -0.072221 , -0.40647 , 0.26899001, -0.28318
, 0.72409999, 0.50796002, -0.37845999, -0.13008 , -0.13808 , 0.098928 , 0.16215999, 0.16293]

Word geometry

Vector Math



Embeddings encode societal bias

$\phi(\text{scientist}) - \phi(\text{woman}) + \phi(\text{man})$:

geologist
engineer
astronomer
mathematician
science

$\phi(\text{scientist}) - \phi(\text{man}) + \phi(\text{woman})$:

anthropologist
sociologist
psychologist
geneticist
biochemist

Embeddings encode societal bias

$\phi(\text{smart}) - \phi(\text{girl}) + \phi(\text{boy})$:

wise
better
guy
kind
good
kid

$\phi(\text{smart}) - \phi(\text{boy}) + \phi(\text{girl})$:

sexy
pretty
incredibly
cute
exciting
funny

Bias in LFW dataset

The screenshot displays the FiftyOne web application interface. At the top, the 'FiftyOne' logo and 'f1w' tag are visible. Below the navigation bar (Samples, Tags, Labels, Scalars), a 'SortBy' dropdown is set to 'uniqueness', and a 'False' filter is applied. The left sidebar shows 'Display Options' and a list of 'TAGS' (No options available). Under 'LABELS', 'ground_truth' is selected with 13,233 samples. Under 'SCALARS', 'uniqueness' is selected with 13,233 samples. The main area shows a grid of 15 face images, each with a label and a uniqueness score. The images are sorted by uniqueness, showing duplicates of the same person (e.g., George W. Bush, Arl Fetscher, Bart Hendricks, Ricky Ray, Julie Gerberding, Carlos Beltran, George W. Bush, Serena Williams, Leslie Ann Woodward) and some images with low uniqueness scores (e.g., 0.202, 0.206, 0.245, 0.246, 0.249, 0.251, 0.258, 0.261, 0.283, 0.283, 0.306, 0.308, 0.322, 0.323, 0.329). The text '0 samples selected' and 'Viewing 13,233 samples' are also visible.

Image	Label	Uniqueness Score
[Image 1]	Arl_Fetscher	0.202
[Image 2]	Arl_Fetscher	0.206
[Image 3]	Bart_Hendricks	0.245
[Image 4]	Ricky_Ray	0.246
[Image 5]	Julie_Gerberding	0.249
[Image 6]	George_W_Bush	0.251
[Image 7]	George_W_Bush	0.258
[Image 8]	Julie_Gerberding	0.261
[Image 9]	Raul_Ibanez	0.283
[Image 10]	Carlos_Beltran	0.283
[Image 11]	George_W_Bush	0.306
[Image 12]	George_W_Bush	0.308
[Image 13]	Serena_Williams	0.322
[Image 14]	Serena_Williams	0.323
[Image 15]	Leslie_Ann_Woodward	0.329

Sorting by the least unique images to find duplicates and incorrect labels

Hacking AI systems

ars TECHNICA

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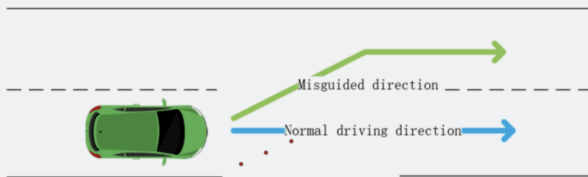
TESLA AUTOPILOT —

Researchers trick Tesla Autopilot into steering into oncoming traffic

Stickers that are invisible to drivers and fool autopilot.

DAN GOODIN - 4/1/2019, 8:50 PM

Keen Security Lab



Machine learning at a large Internet company

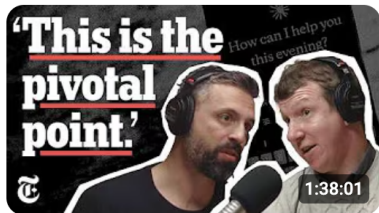
- Typical project lifetime: 6 months to 1 year
- Ads projects involve thousands of software engineers
- Often adding new “feature” to existing black box model
- No single person understands entire model
- Not interpretable
- Security issues with data

Questions of the day



Brian Eno on AI, Art and What Makes Us Human ⋮

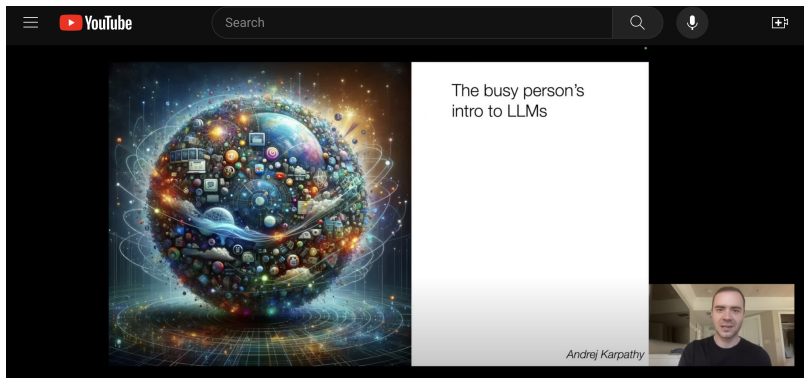
The Ezra Klein Show ✓ and Brian Eno 🎧
165K views · 6 months ago



How Fast Will A.I. Agents Rip Through the Economy? | The ... ⋮

The Ezra Klein Show ✓ and 2 more
352K views · 1 month ago

Recommended viewing



https://www.youtube.com/watch?v=zjkBMFhNj_g

Recommended viewing

LLM Security



https://www.youtube.com/watch?v=zjkBMFhNj_g&t=45m43s

Jailbreaks: https://www.youtube.com/watch?v=zjkBMFhNj_g&t=46m15s

Prompt injection: https://www.youtube.com/watch?v=zjkBMFhNj_g&t=51m30s

Data poisoning: https://www.youtube.com/watch?v=zjkBMFhNj_g&t=56m23s

Panel Discussion



Rafael Leifert



Yuki Sugita



Matt Sareno



Yurii Stasiuk



Maneet Mehta



Frank Lin



Frank Liu